FP8 Dispatch + BF16 Combine

Ship each MoE token in the precision its receiver uses

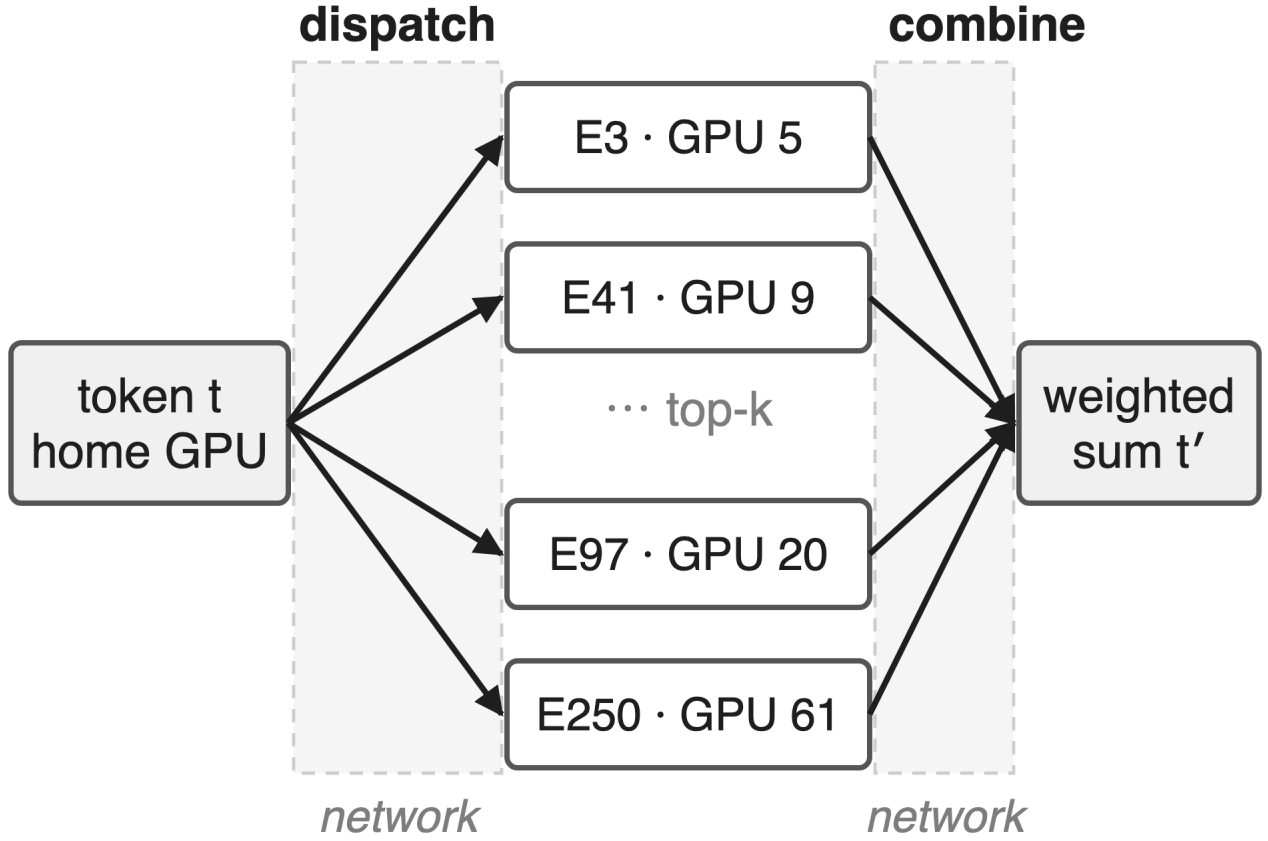
DeepSeek-AI · DeepEP, DeepSeek-V3 and V4

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

Expert-Parallel Communication · github.com/deepseek-ai/DeepEP

Every MoE token crosses the network twice

- DeepSeek-V3: **256** experts, top-**8**, h = **7168**
- **2048** H800: IB **50** GB/s vs NVLink **160** GB/s
- Dispatch copies a token to k experts; combine sums k results



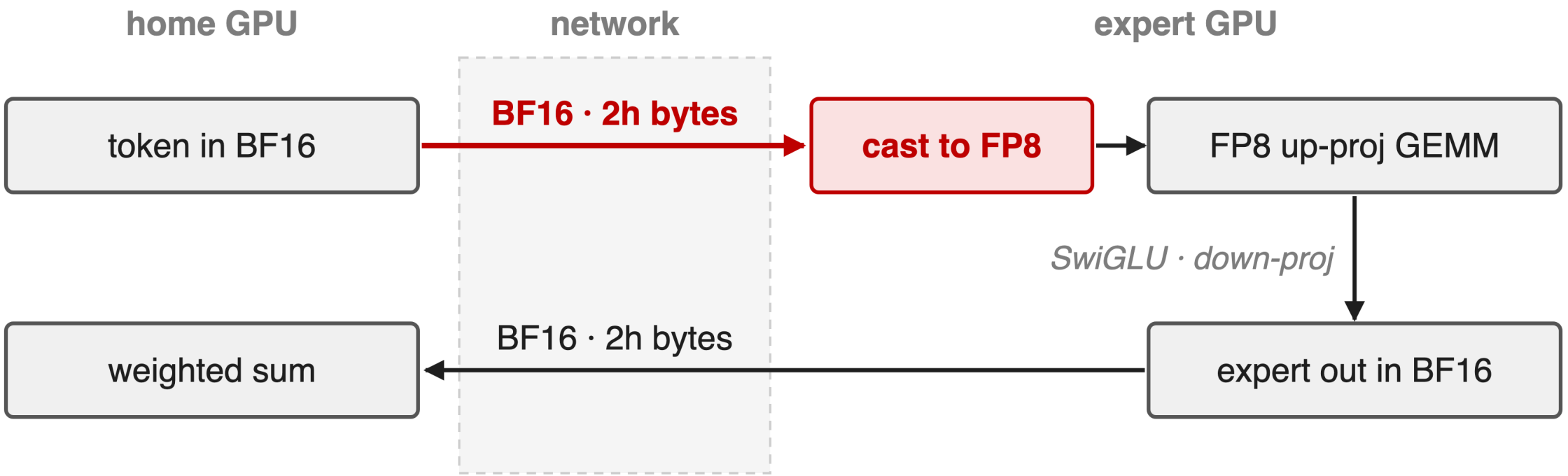
Cross-node EP makes V3 communication-bound

≈ 1:1

computation-to-communication ratio of DeepSeek-V3 training with cross-node expert parallelism

DeepSeek-V3 Technical Report, arXiv 2412.19437, §3.2.1 (p.12)

BF16 dispatch ships bits the receiver discards



Half of every dispatched byte is precision the FP8 GEMM never reads

Insight: dispatch feeds an FP8 GEMM anyway

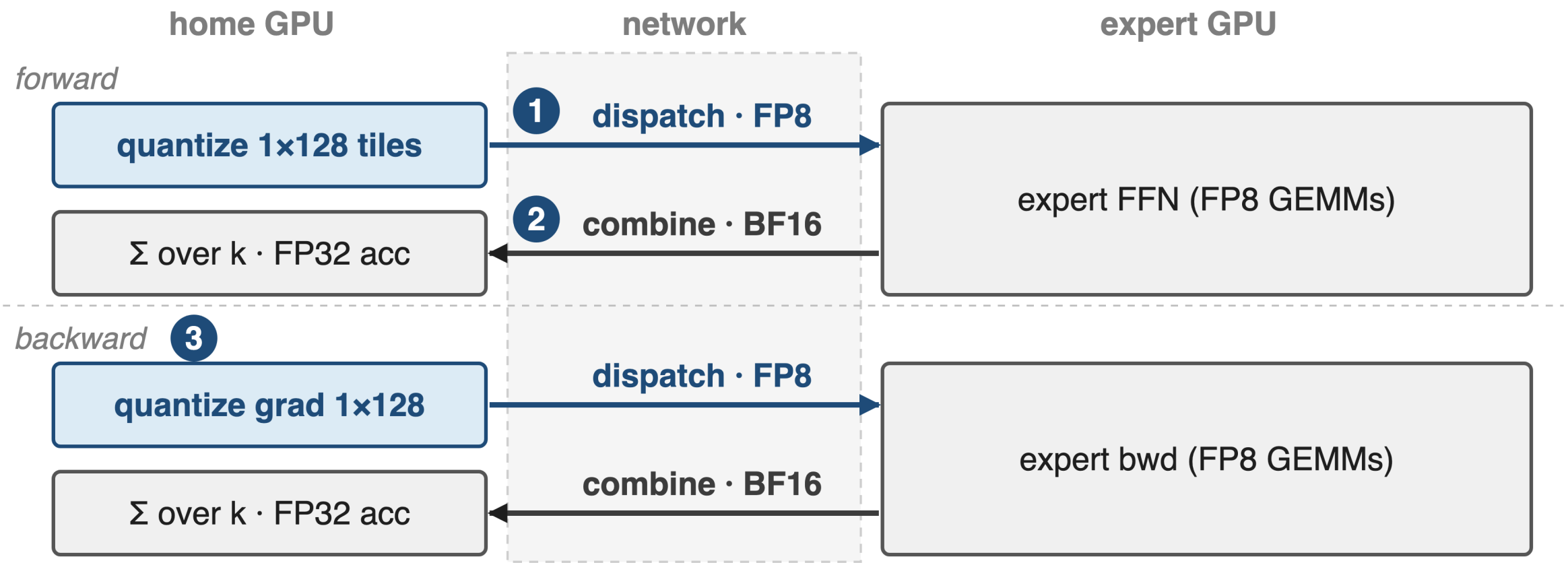
- Forward: up-projection runs FP8 on dispatched tokens
- Backward: down-projection gradient GEMM consumes FP8 too
- Combine output is summed into the residual stream

Insight: dispatch feeds an FP8 GEMM anyway

- Forward: up-projection runs FP8 on dispatched tokens
- Backward: down-projection gradient GEMM consumes FP8 too
- Combine output is summed into the residual stream

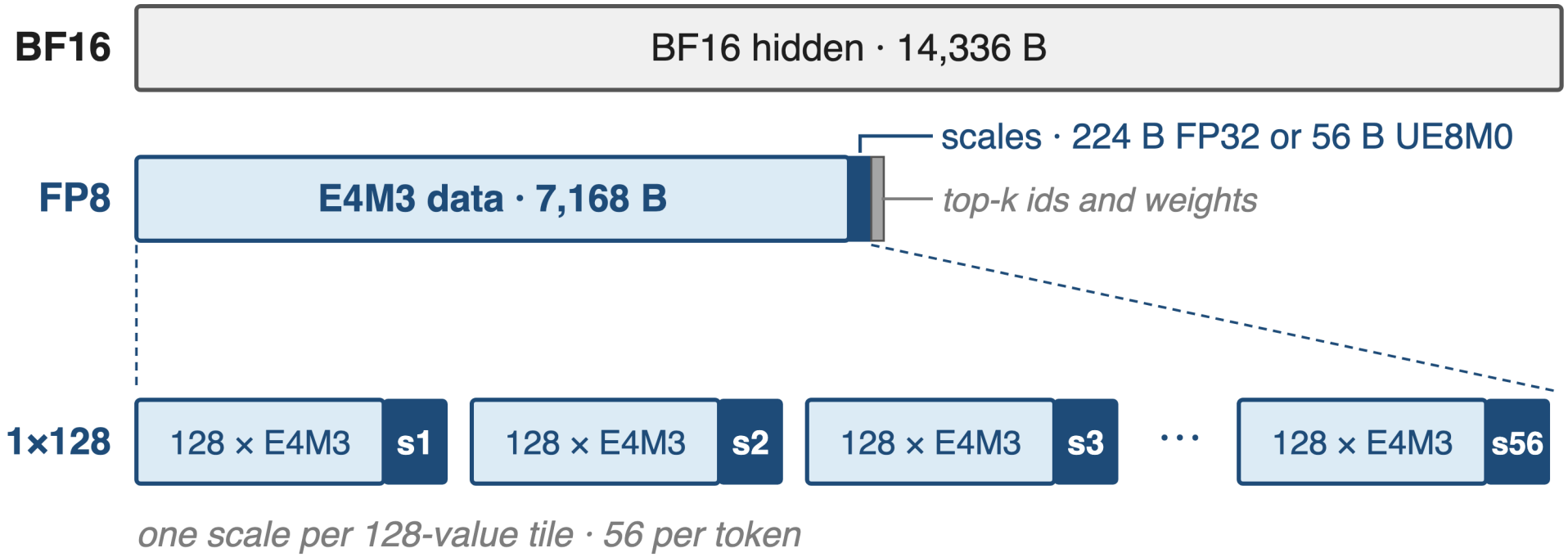
Quantize before the wire where the receiver would; keep BF16 where it sums

Precision follows the direction of traffic



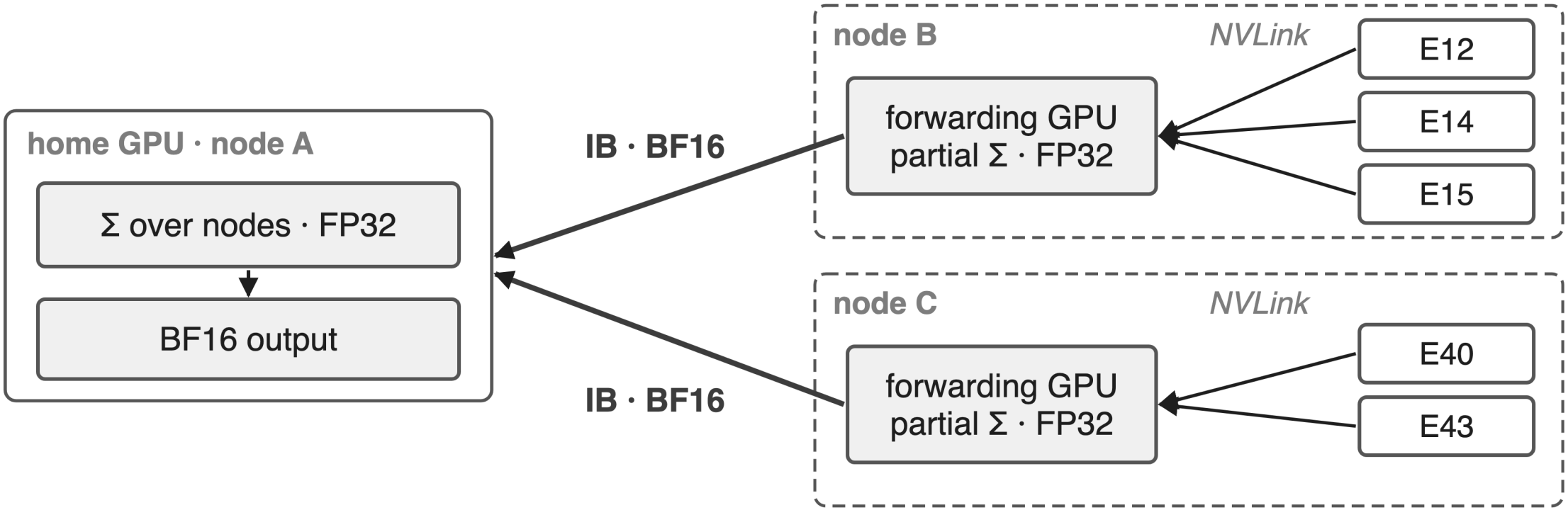
Token→expert: FP8 with 1×128 scales (❶). Expert→token: BF16, summed in FP32 (❷). Backward mirrors forward (❸).

❶ Quantize per 1×128 tile, ship scales inline



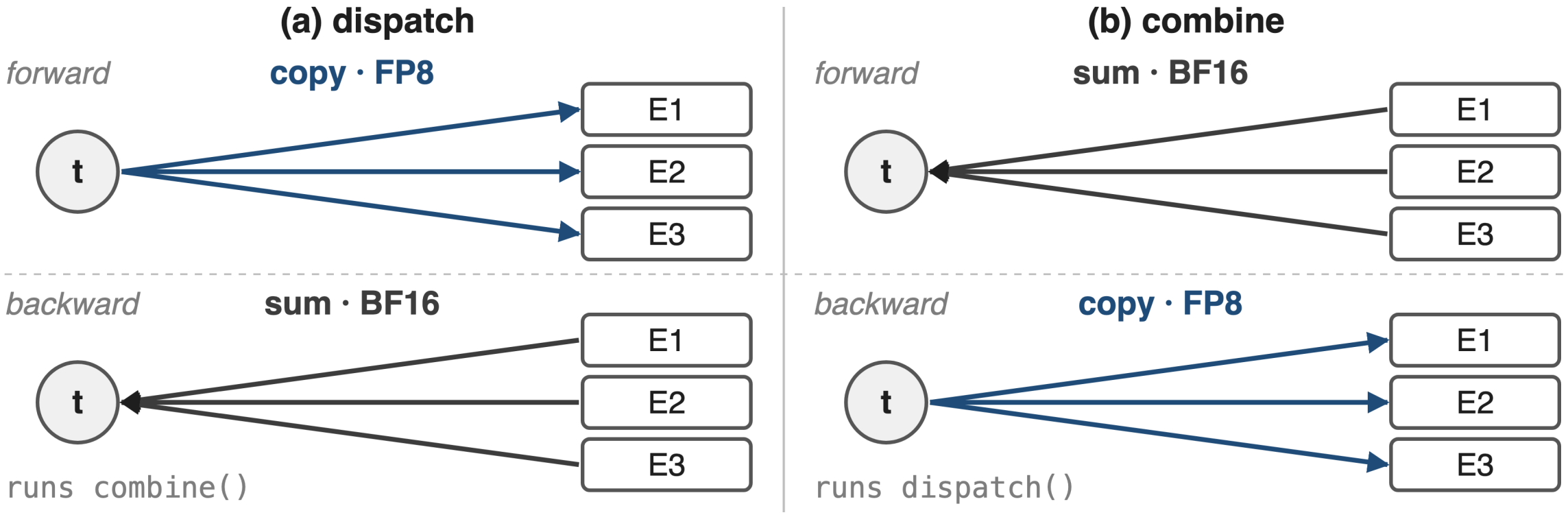
Scales travel with the data; the receiver lays them out for the grouped GEMM

❷ Combine stays BF16 and sums in FP32



Every sum accumulates in FP32; only BF16 crosses the wire

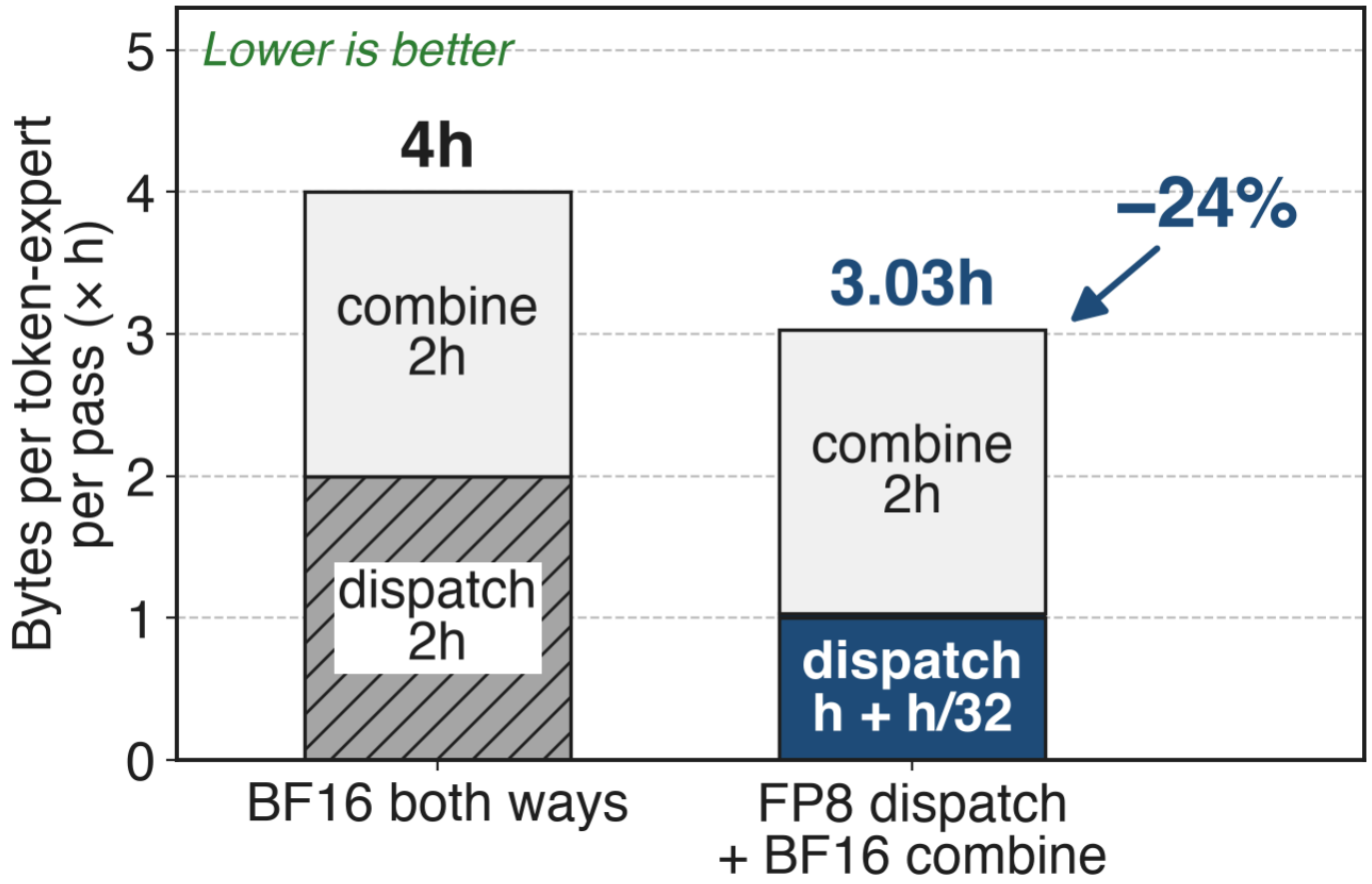
❸ Backward reuses both kernels, mirrored



Token→expert traffic is always FP8; expert→token traffic is always BF16

FP8 dispatch moves 3h instead of 4h bytes

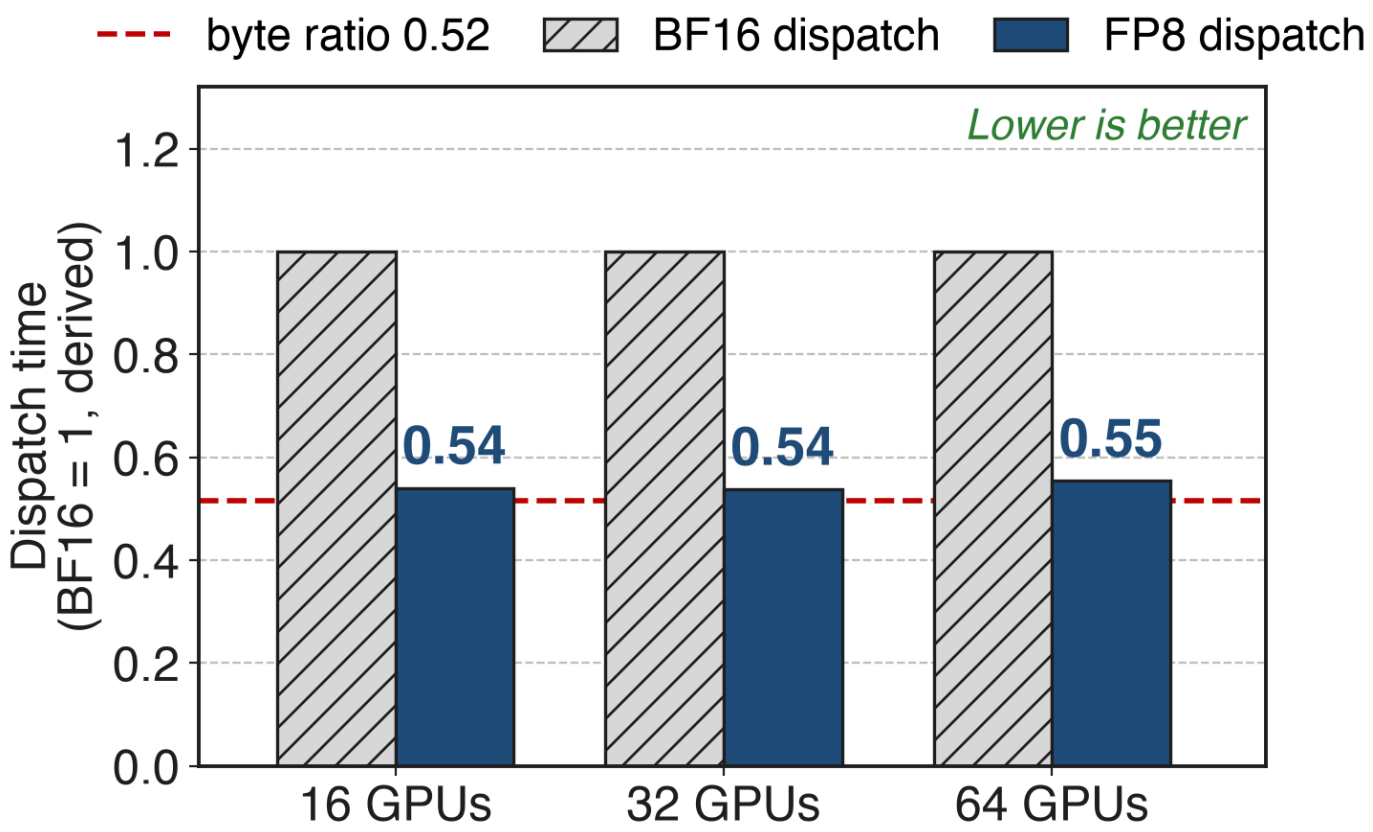
- h = 7168: **28.7** KB → **21.7** KB per pair
- Dispatch alone: about **half** the bytes
- Backward saves the same share



Analytic; FP32 scales per 128 channels (UE8M0: 3.008h); top-k metadata excluded

FP8 dispatch finishes in 54–55% of BF16 time

- Hybrid-EP kernels on H100 over IB, **16** SMs
- **4096** tokens, h **7168**, top-**8**
- Time tracks bytes: the link stays saturated



Derived: published GB/s × the benchmark's FP8 byte count (0.516 of BF16)

Fewer bytes raise V4's balance point to 6144

6144

FLOPs per byte: 6hd FLOPs vs 3h bytes per token-expert, so EP hides when C/B ≤ 2d (4608 if BF16 both ways, derived)

DeepSeek-V4, arXiv 2606.19348, §3.1 "Observations and Proposals" (p.16); V4-Pro d = 3072 (§4.2.1)

The cost: a cast, scales, and BF16 combine

Costs

- Quantize step; scales add **3%** of h
- Combine still carries **2/3** of the bytes
- Needs FP8 expert GEMMs and recipe

Open fronts

- Low-bit combine: LogFMT codec costs **50–100%**
- NVFP4: listed for the Hybrid-EP branch

Summary

- Cross-node EP: compute to communication ≈ **1:1** in V3 training
- **Insight**: dispatch feeds an FP8 GEMM; combine is a sum
- **3h** not **4h** bytes per pair; FP8 dispatch in **~55%** of BF16 time
- V4 balance point: **6144** FLOPs/Byte

https://github.com/deepseek-ai/DeepEP